HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F78PB

Probability and Statistics B

Semester 2 2020/21

Duration: Two Hours

ANSWER ALL QUESTIONS

Total Marks: 60

Note: To receive full credit, you must show your workings and explain your answers. A correct answer with no explanation may gain no marks.

1. It is claimed that the way people use social media is changing with more people now taking part in gaming activities than engaging with platforms like Facebook or Instagram.

A random sample of 20 university students was asked how many hours they spent on gaming in the first week of April 2020. The results are shown below, listed in increasing order.

0 0 4.0 4.7 5.8 6.0 6.0 6.4 6.7 7.2 7.2 7.2 8.6 9.2 10.2 10.4 10.8 11.2 11.4 21.0

Note that $\sum x = 154$ and $\sum x^2 = 1568.94$

(a) Calculate the mean and standard deviation of the number of hours spent on gaming.

[2 marks]

(b) Calculate the median and interquartile range of the number of hours spent on gaming.

[3 marks]

(c) Another random sample of 20 university students was asked how many hours they spent on Facebook and Instagram in the first week of April 2020 and a summary of results is shown below.

Minimum	1st Quartile	Median	3rd Quartile	Maximum
0	4.8	6.2	8.6	12.6

Draw two boxplots side by side for gaming and Facebook/Instagram and comment on them.

[4 marks]

[Total 9 marks]

2. (a) Let $\underline{X}$ denote a random sample of size n from a distribution with parameter θ and let $g(\underline{X})$ denote an estimator for θ.

(i) Show that $MSE(g(\underline{X})) = Var(g(\underline{X})) + (bias\ (g(\underline{X})))^2$.

[2 marks]

(ii) Why can it be helpful to express $MSE(g(\underline{X}))$ in this way?

[1 mark]

(b) Let $\underline{X} = (X_1, X_2, \ldots\ldots, X_n)$ be a random sample of size $n \geq 2$ from a Poisson distribution with parameter λ.

Two possible estimators of λ are suggested by $\hat{\lambda}_1 = \frac{X_1 + X_2}{2}$ and $\hat{\lambda}_2 = \bar{X}$ (the sample mean).

Note: For this question you can make use of the distributions formula sheet.

(i) Calculate the bias and variance of each estimator.

[4 marks]

(ii) Calculate the Mean Square Error (MSE) of each estimator and determine which of these two estimators is more efficient.

[3 marks]

(iii) Show that $\hat{\lambda}_2$ is the method of moments estimator (MME) of λ.

[2 marks]

[Total 12 marks]

3. A new drug is being trialled to help reduce heartburn in people over the age of 60.

Nine individuals in this age-group who had similar health profiles were tested on a new pill to be taken before bedtime. The time until each person's next bout of heartburn, in days, is thought to be modelled by an Exp (β) distribution where β is an unknown parameter.

All 9 individuals were given the drug on the same day at the same time and each is monitored over the following 10 days. During this time, 3 individuals have a bout of heartburn after taking the drug at times (t) 3.7, 5.1 and 7.2 days. The remaining 6 individuals did not suffer heartburn at all during the monitoring period.

(a) Show carefully that the likelihood for these data is given by $L(\beta) = \beta^3 e^{-76\beta}$.

[4 marks]

(b) Find an equation for $\hat{\beta}$, the maximum likelihood estimate (MLE) of β and use it to estimate β. Show all steps involved in finding the equation for $\hat{\beta}$.

[4 marks]

(c) Give a reason why a statistical censoring technique like this might be used.

[1 mark]

[Total 9 marks]

4. For an investigation into alcohol drinking habits, a random sample of men aged 16 to 90 is obtained. The following data are reported for men belonging to two different age groups:

Age Group	16 - 44	45 - 90
Average units per week	4.2	4.9
Sample standard deviation	1.6	1.4
Sample size	50	60

(a) Calculate a 99% confidence interval for the expected units of alcohol per week consumed by men aged 16 – 44 based on the sample above.

[2 marks]

(b) Test the hypothesis that the mean number of units of alcohol consumed per week is the same for both age groups.

[5 marks]

(c) For both (a) and (b) what assumptions are you making about the distribution of average units per week consumed by the men?

[1 mark]

[Total 8 marks]

5. A supermarket chain claims that breakfast cereal eaters are not able to tell the difference between their own (cheaper) brand of cornflakes and a leading major brand.

To test the hypothesis that people cannot tell the difference, a statistician organises a tasting experiment where an individual is asked to taste 10 randomly presented bowls of cornflakes; five of these being the supermarket's own brand and the other five the leading brand. The individual does not know how many cornflake samples were from either brand.

Under the null hypothesis that people cannot tell the difference, the probability of correctly identifying the brand of cornflakes purely by chance is considered to be 0.5. A one-sided test is used with H_0: $p = 0.5$ versus H_1: $p > 0.5$.

The statistician adopts the following decision rule for testing the hypothesis:

- Reject the null hypothesis if seven or more bowls of cornflakes are correctly identified; conclude that the individual can tell the difference.
- Do not reject the null hypothesis if less than seven bowls of cornflakes are correctly identified; conclude that the individual cannot tell the difference.

(a) Determine the probability of a type I error for this test.

[3 marks]

(b) Based on past experience we can quantify the probability p that an individual recognises the brand of cornflakes correctly. Determine the probability of a type II error assuming that the alternative hypothesis is of the form

(i) H_1: $p = 0.7$.

[2 marks]

(ii) H_1: $p = 0.8$.

[2 marks]

(c) Calculate the power of the test for each of your answers in part (b) and comment on how this changes as p increases.

[3 marks]

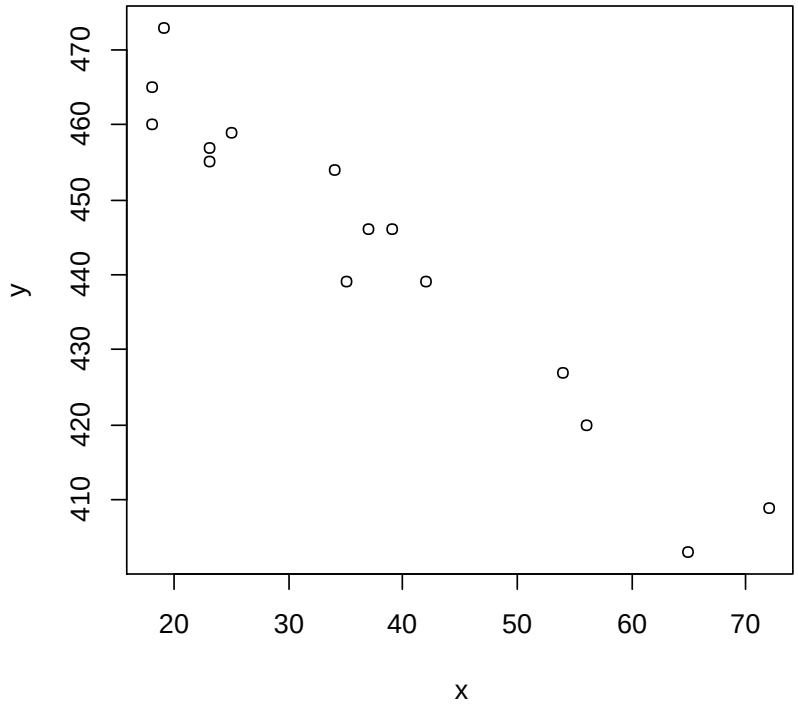
[Total 10 marks]

6. A recent study suggests that the sleeping time, in minutes, (y) required by adults according to their age, in years (x) is given by the equation, Sleeping Time = 480 – Age.

Suppose this is to be empirically checked and 15 people of varying ages are asked to report how long they slept on a particular standard night. The following data are collected:

x	18	23	25	35	65	54	34	56	72	19	23	42	18	39	37
y	460	457	459	439	403	427	454	420	409	473	455	439	465	446	446

(a) By referring to the scatter diagram below, comment on the relationship between age and sleeping time.



[2 marks]

(b) Given that

$\sum x = 560,\ \sum y = 6652,\ \sum x^2 = 25188,\ \sum y^2 = 2955918,\ \sum xy = 243444,\ \hat{\sigma}^2 = 28.906,$

calculate the fitted line for the regression of y on x. Show your working.

[3 marks]

(c) Assuming that we are dealing with a full normal model of regression (in other words the errors are independently and identically distributed as Normal with a mean of 0 and some variance σ^2), find a 95% confidence interval for an individual value for the number of hours sleep taken by a 60 year old person. Show your working.

[3 marks]

(d) Does your answer to (c) seem to support the claim of the study that Sleeping Time = 480 – Age? Give a reason.

[2 marks]

(e) In the standard regression model, why are the predicted errors different for expected Y at X and for individual Y at X? (refer to the formula sheet).

[2 marks]

[Total 12 marks]

[Total exam marks 60]

END OF EXAMINATION